

Coordination-Induced Conformational Control Enables Highly Luminescent Metallo-Cages

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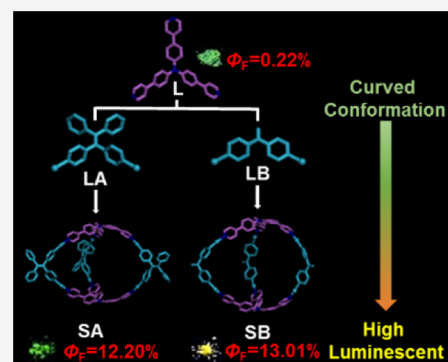


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Supporting Information

ABSTRACT: In recent years, luminescent materials have received a great deal of attention due to their wide range of applications. However, exploring a simple solution to overcome the fluorescence quenching resulting from the aggregation of conventional organic fluorophores remains a valuable area of investigation. In this study, we successfully constructed two metallo-cages, namely, SA and SB, through coordination-driven self-assemblies of the triphenylamine (TPA)-based donor L with different diplatinum(II) acceptors LA and LB, respectively. These metallo-cages take advantage of their steric nature and curved conformation to more effectively limit the free rotation of the benzene ring and hinder π – π stacking in the solid state, which successfully inhibited fluorescence quenching and realizing highly efficient luminescent properties. Therefore, this work offers a new design strategy for preparing materials with excellent luminescent properties.



INTRODUCTION

In the past few decades, the design and synthesis of luminescent materials have received significant attention due to their widespread applications in fields such as sensing,^{1,2} fluorescent probes,^{3,4} and lighting devices^{5,6} and other fields.^{7,8} However, the vast majority of conventional organic fluorophores suffer from aggregation-caused quenching (ACQ) in concentrated solution or solid state,^{9,10} such as triphenylamine (TPA),¹¹ anthracene,^{12,13} pyrene,^{14,15} peryleneimine (PDI),¹⁶ etc., limiting their practical applications. To address this issue, in 2001, Tang et al. developed aggregation-induced emission (AIE) fluorophores, which can enhance luminescence in the aggregation state.¹⁷ However, AIE-active fluorophores generally face difficulties in emitting intense fluorescence in the dilute solution state due to high nonradiative decay caused by the free rotation of the benzene ring.^{18,19} More importantly, the variety of AIE fluorophores is very limited, so scientists are still trying various methods to avoid the ACQ behavior of the traditional chromophore, for example, the introduction of large steric hindrance groups,^{20,21} covalent doping,^{5,22} wrapping and adsorption of other particles,²³ and so on. Although these methods reduce the impact of aggregation, they still have disadvantages such as complex synthesis, low yield, and destruction of the original functionality of the material. Therefore, it is crucial to find a simple and efficient method to inhibit molecular aggregation and achieve efficient luminescence in both solution and solid state simultaneously.

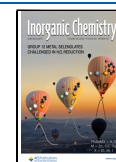
Coordination-driven self-assembly is a powerful and straightforward approach for synthesizing supramolecular architectures because of the definite directionality, predictable

nature of metal–ligand coordination interactions, as well as moderate bond energies (15–25 kcal/mol) of the metal–ligand coordination interactions.^{24,25} Therefore, a series of discrete emissive metal–organic complexes (MOCs) with well-defined sizes and shapes, ranging from metallo-macrocycles^{26–29} to metallo-cages^{30–33} have been successfully prepared by introducing different chromophores into pre-designed ligands. Among these MOCs, metallo-cages exhibited greater degrees of restricted free rotation of chemical groups and are better at exploiting their luminescence properties than metallo-macrocycles.³⁴

Triphenylamine (TPA), as a classical fluorescent motif, has been extensively studied for its electrically rich properties and high hole mobility, but its aggregation-caused quenching (ACQ) issues limit its practical application.^{35,36} Herein, we report the construction of two metallo-cages SA and SB with TPA motif through the self-assembly of TPA-based donor L with different di-platinum(II) acceptors LA and LB, respectively. The presence of coordination bonds resulted in a certain bending of the triphenylamine core, thereby confining rotational movement of the benzene ring and facilitating high-efficiency luminescence under dilute solution ($\Phi_F = 31.99\%$ for SA, $\Phi_F = 53.89\%$ for SB). Furthermore, compared with the

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Scheme 1. Self-Assembly of Metallo-Cages SA and SB (Energy-Minimized Structures)

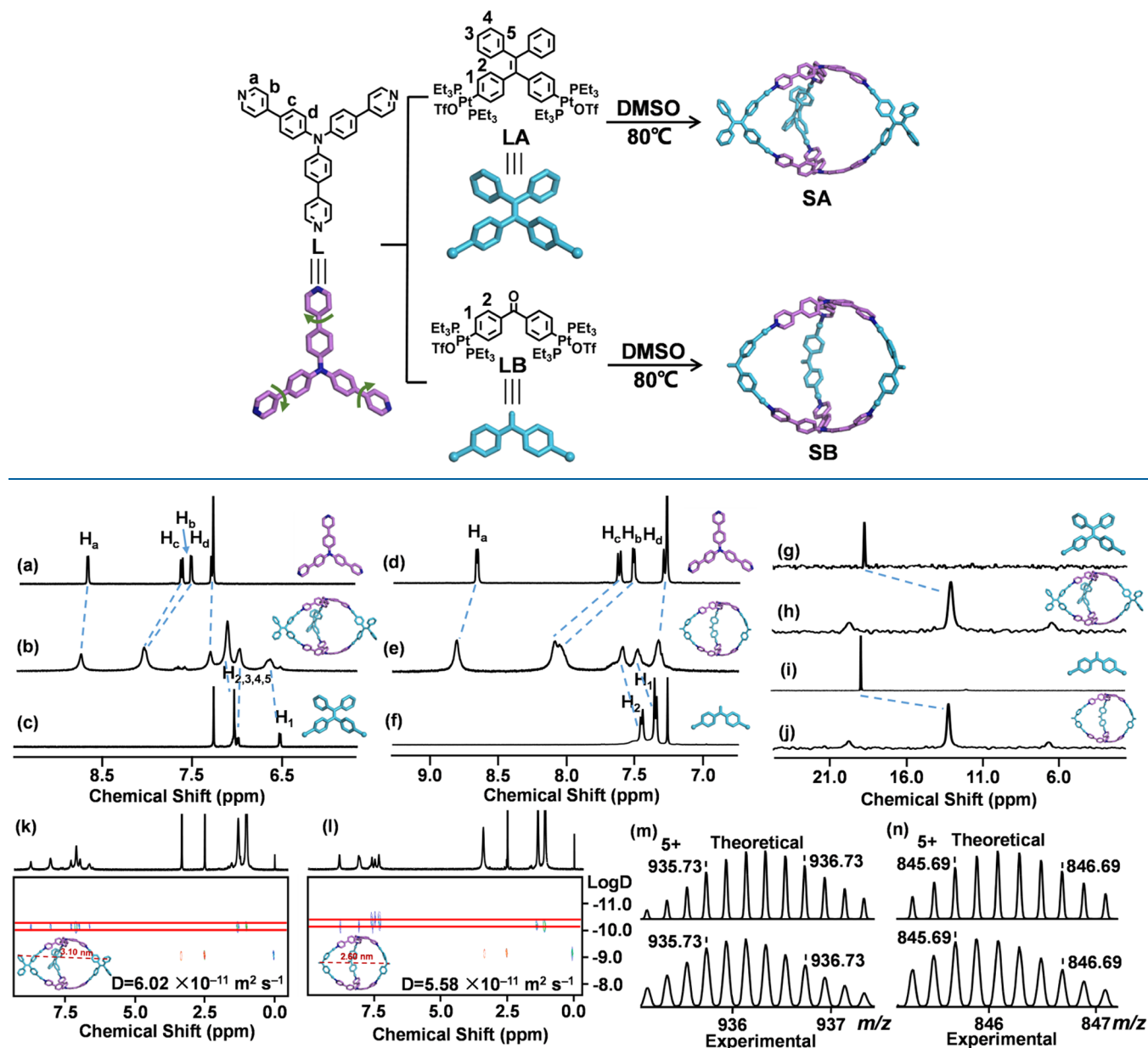


Figure 1. ^1H NMR spectra (500 MHz, 300 K) of **L** (a and d), **LA** (c), and **LB** (f) in CDCl_3 , **SA** (b) and **SB** (e) in $\text{DMSO}-d_6$, ^{31}P NMR spectra (500 MHz, 300 K) of **LA** (g) and **LB** (i) in CDCl_3 and **SA** (h) and **SB** (j) in $\text{DMSO}-d_6$. 2D DOSY (500 MHz, $\text{DMSO}-d_6$, 300 K) spectrum of **SA** (k) and **SB** (l). ESI-MS spectra of **SA** (m) and **SB** (n).

TPA-based ligand **L**, the twisted conformation of these metallo-cages hindered the π – π stacking in the aggregation state, effectively suppressing the ACQ behavior and achieving enhanced luminescence efficiency.

EXPERIMENTAL SECTION

Compounds **LA** and **LB** were prepared according to the reported literature.^{37,38} Column chromatography was conducted using basic SiO_2 (VWR, 40–60 μm , 60 Å), and the separated products were visualized by UV light. NMR spectra data was recorded on Bruker Avance 400 MHz, 500 MHz, and 600 MHz NMR spectrometer in CDCl_3 and $\text{DMSO}-d_6$ with TMS standard as the reference. Matrix-assisted laser desorption/ionization time of flight mass spectrometry (MALDI-TOF-MS) was performed on a Bruker AutoflexIII instrument using *trans*-2-[3-(4-*tert*-butylphenyl)-2-methyl-2-propenyldene] malononitrile (DCTB) as a matrix. ESI-MS was recorded

with a Waters Synapt G2 tandem mass spectrometer, using solutions of 0.5 mg of sample in 1 mL of $\text{DMSO}/\text{MeOH}/\text{acetone}$ (1:1:3, v/v) for complexes. The UV–vis spectra of the solution samples were recorded with a Shimadzu UV2550 spectrophotometer. The emission spectra of solution samples were measured on a Shimadzu RF-5301 PC spectrometer (CCD) and a Maya2000Pro optical fiber spectrophotometer. The UV–vis spectra of the solid samples were recorded with a PerkinElmer Lambda1050+ spectrophotometer. The emission spectra of solid samples were determined by an Edinburgh FLS920 steady-state and transient-state fluorescence spectrometer. The solid-state and solution-state quantum yields were recorded by an Edinburgh FLS980-S2S2-stm steady-state and transient-state fluorescence spectrometer.

Synthesis of Ligand L. Tris(4-bromophenyl)amine (3.0 g, 6.27 mmol), 4-pyridylboronic acid (3.0 g, 24.43 mmol), $\text{Pd}(\text{PPh}_3)_2\text{Cl}_2$ (658 mg, 0.94 mmol), and Cs_2CO_3 (12.25 g, 37.59 mmol) were added into a 200 mL Schlenk flask. After evacuation under vacuum

and flushing with nitrogen three times, dioxane (80 mL) and H₂O (20 mL) were injected into a Schlenk flask. Then, the mixture was stirred at 88 °C for 3 days. After cooling to room temperature, the mixture was extracted with dichloromethane and the crude product was purified by flash column chromatography with dichloromethane/ethanol (100:3.0, v/v) and was given as a yellow solid (0.18 g, 80%). ¹H NMR (400 MHz, CDCl₃) δ (ppm): 8.66 (d, *J* = 4.6 Hz, 6H, Ph-H_a), 7.61 (d, *J* = 8.4 Hz, 6H, Ph-H_c), 7.51 (d, *J* = 4.7 Hz, 6H, Ph-H_b), 7.28 (d, *J* = 8.8 Hz, 6H, Ph-H_d). ¹³C NMR (100 MHz, CDCl₃) δ (ppm): 150.42, 147.98, 147.50, 133.02, 128.20, 124.83, 121.17, 77.48, 77.16, 76.84. MALDI-TOF: calcd. for [C₃₃H₂₄N₄+H]⁺ 476.2001 (found 476.2025).

Synthesis of Metallo-Cage SA. Ligands **L** (1.0 mg, 2.1 μmol) and **LA** (4.7 mg, 3.2 μmol) were dissolved in 0.8 mL of DMSO. The mixture was heated at 80 °C for 10 h. After cooling to room temperature, 5.0 mL of ethyl ether was used to precipitate the product. After the mixture was filtered and washed with diethyl ether, pure **SA** was obtained as a yellow solid (5.4 mg, 94.7%). ¹H NMR (500 MHz, DMSO-*d*₆, 300 K) δ (ppm): 8.73 (s, 12H, Ph-H_a), 8.03 (s, 24H, Ph-H_b, Ph-H_c), 7.29 (s, 12H, Ph-H_d), 7.04 (d, *J* = 66.7 Hz, 42H, Ph-H₂, Ph-H₃, Ph-H₄, Ph-H₅), 6.64 (s, 12H, Ph-H₁). ¹³C NMR (150 MHz, DMSO-*d*₆) δ (ppm): 152.30, 134.97, 132.03, 131.49, 130.74, 130.35, 128.81, 127.65, 124.74, 123.51, 122.80, 121.74, 121.61, 121.24, 119.60, 39.80, 39.66, 39.52, 39.38, 39.24, 11.97, 7.30. ESI-MS (*m/z*): 935.73 [M - SOTf]⁵⁺ (calcd *m/z*: 935.73).

Synthesis of Metallo-Cage SB. Ligands **L** (1.1 mg, 2.3 μmol) and **LB** (4.6 mg, 3.5 μmol) were dissolved in 0.8 mL of DMSO. The mixture was heated at 80 °C for 10 h. After cooling to room temperature, 5.0 mL of ethyl ether was used to precipitate the product. After filtering and washing with diethyl ether, pure **SB** was obtained as an orange solid (5.4 mg, 95%). ¹H NMR (500 MHz, DMSO-*d*₆, 300 K) δ (ppm): 8.81 (s, 12H, Ph-H_a), 8.14–7.99 (m, 24H, Ph-H_b, Ph-H_c), 7.59 (s, 12H, Ph-H₂), 7.48 (s, 12H, Ph-H₁), 7.33 (s, 12H, Ph-H_d). ¹³C NMR (150 MHz, DMSO-*d*₆) δ (ppm): 152.38, 143.95, 136.02, 132.29, 132.04, 131.50, 128.85, 124.78, 123.64, 119.60, 39.80, 39.66, 39.52, 39.38, 39.24, 11.99, 7.34. ESI-MS (*m/z*): 845.69 [M - SOTf]⁵⁺ (calcd *m/z*: 845.69).

■ RESULT AND DISCUSSION

The synthesis and characterization of the 120° di-Pt(II) acceptors **LA**, **LB**, and the TPA-based donor **L** are shown in the Supporting Information (Figures S1–S8). As shown in Scheme 1, the obtained TPA-based donor **L** was mixed with 120° di-Pt(II)-based acceptors **LA** and **LB** in a 2:3 stoichiometric ratio, respectively. The mixtures were dissolved in DMSO and heated at 80 °C for 10 h, followed by the addition of an excess of diethyl ether to give yellow and orange precipitates (yield 94.7% for complex **SA**, yield 95.0% for complex **SB**).

The formations of metallo-cages **SA** and **SB** were first confirmed by multinuclear NMR analysis [¹H NMR, ³¹P NMR, and ¹³C NMR] (Figures 1a–1j, S9–S11, and S15–S17). As shown in Figure 1, compared to their ligands, complexes **SA** and **SB** showed broadened signals, suggesting the limited free rotation of phenyl groups after coordination. The signals of protons corresponding to the pyridyl units and phenyl rings in metallo-cage **SA** (Figure 1b) showed downfield shifts relative to those of the free ligand **L**, indicating the formation of Pt–N coordination bonds because of the loss of electron density. A similar downfield shift was also observed in the spectra of ligand **L** and metallo-cage **SB** (Figure 1d,e). All peaks were accurately assigned by 2D COSY and 2D NOESY spectra (Figures S12, S13, S18, and S19). In addition, the ³¹P NMR spectrum of both metallo-cages **SA** and **SB** displayed a sharp singlet with concomitant ¹⁹⁵Pt satellites (δ = 13.09 ppm for **SA** and 13.25 ppm for **SB**, respectively) (Figure 1g–1j),

which was upshifted about 5.66 and 5.74 ppm from the original di-platinum acceptors **LA** and **LB**, respectively. This further demonstrated the successful preparation of single-component assembly with high symmetry. 2D diffusion-ordered spectroscopy (DOSY) in DMSO-*d*₆ showed a single band with diffusion coefficients (*D*) of 6.02 × 10^{−11} m² s^{−1} for **SA** and 5.58 × 10^{−11} m² s^{−1} for **SB**, respectively, also indicating the formation of a single-component product (Figure 1k,l). From the log *D* values by the Stokes–Einstein equation,³⁹ the estimated diameter of **SA** and **SB** were 3.3 and 3.5 nm, respectively, which matched well with the sizes of the expected structures. Electrospray ionization mass spectrometry (ESI-MS) was used to further demonstrate the molecular composition of the obtained assemblies.⁴⁰ Isotopically resolved peaks corresponding to an intact entity with charge states arising from the loss of different numbers of OTf[−] counterions were identified for each assembly. For example, a peak at *m/z* = 935.73 Da corresponding to [M - SOTf]⁵⁺ for **SA** (Figure 1m) and *m/z* = 845.69 Da corresponding to [M - SOTf]⁵⁺ for **SB** (Figure 1n) supported their 2:3 stoichiometric ratio.

All attempts to grow X-ray-quality single crystals have been proven to be unsuccessful. Molecular simulations were used to further understand the architectural features of these two metallo-cages (Figure 2). All calculations were carried out with

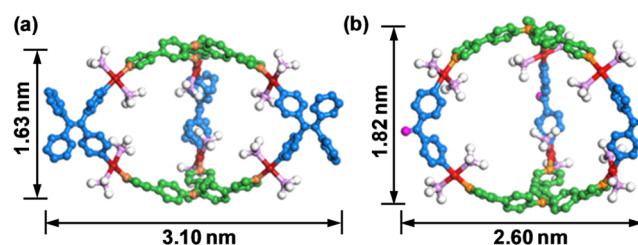


Figure 2. DFT-optimized ground-state structures of **SA** (a) and **SB** (b). The hydrogen atoms, counterions OTf[−] ((trifluoromethanesulfonate) anions), and ethyl groups of the PET₃ were omitted for clarity.

the Gaussian 16 software.⁴¹ Density functional theory (DFT) calculations were carried out with the PBE0 functional with the combination of Grimme's D3 version of dispersion correction.⁴² The effective core potentials with an SDD basis set for Pt and 6-31G for other atoms were adopted for the geometry optimization and frequency calculations. The geometries were fully optimized without any structural constraints in the DMF solution with the SMD solvation model. The harmonic frequency calculations were carried out at the same level of theory to verify that all structures have no imaginary frequency. The calculation model showed that the two complexes are cage structures in which the triphenylamine core undergoes some bending.

After the successful structural characterization of ligands and metallo-cages **SA** and **SB**, their photophysical properties were further studied. As shown in Figure 3, ligand **L** in DMSO exhibited main absorption peaks at 365 nm attributed to the TPA core. The TPE-based di-Pt(II) acceptor **LA** in DMSO exhibited main absorption peaks at 340 nm attributed to the TPE core. The diplatinum receptor **LB** exhibited only a weak absorption band at 315 nm in DMSO. Fluorescence spectra of ligand **L** displayed an emission peak at about 455 nm in DMSO corresponding to the TPA group. The di-Pt(II) acceptors **LA** and **LB** hardly emitted in DMSO due to the nonradiative decay resulting from the free rotation of phenyl

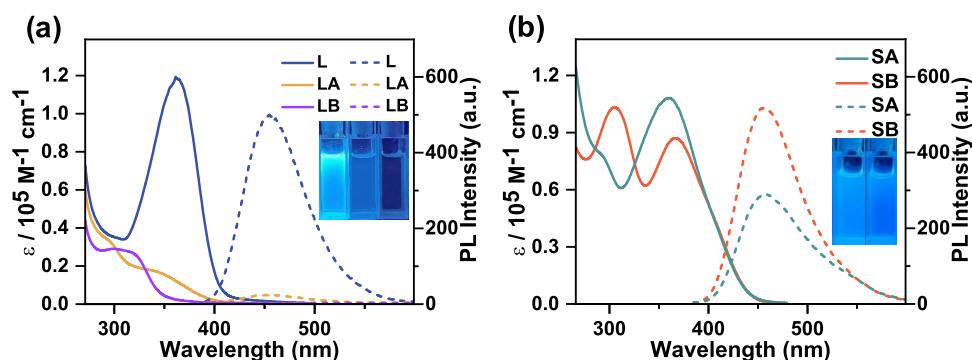


Figure 3. (a) UV-vis and PL spectra of L, LA, and LB in DMSO. (b) UV-vis and PL spectra of SA and SB in DMSO ($\lambda_{\text{ex}} = 365 \text{ nm}$ and $c = 10.0 \mu\text{M}$).

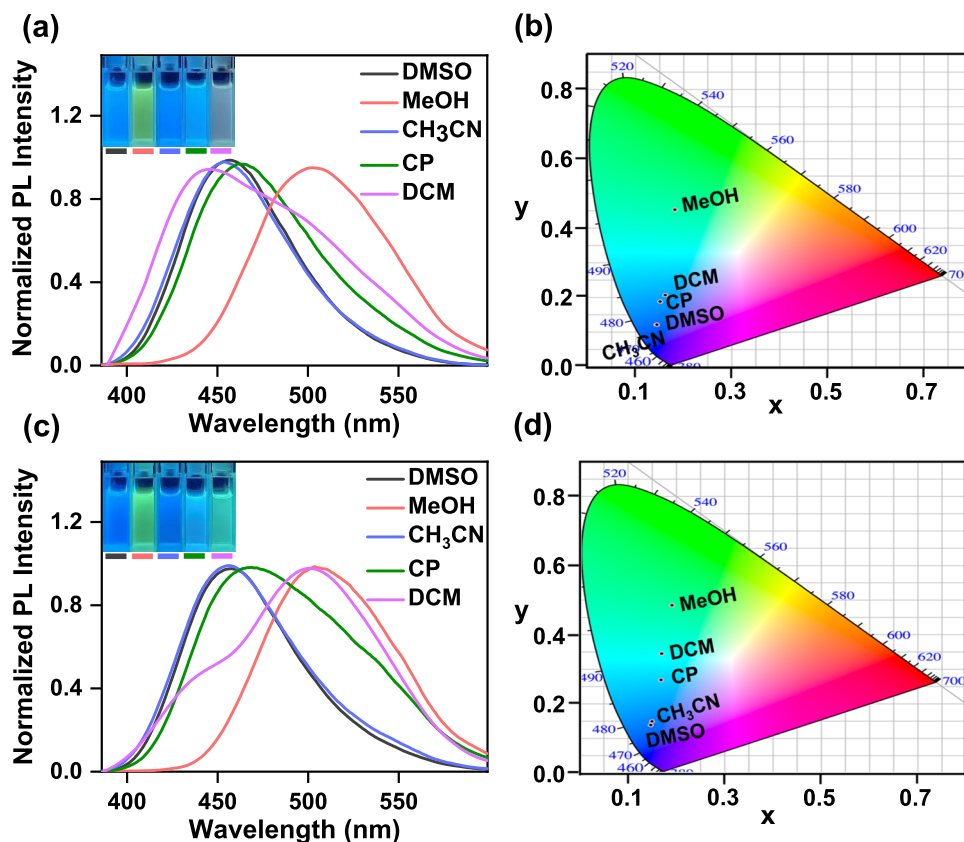


Figure 4. Solvent effect on the light-emitting properties (a) and CIE 1931 chromaticity diagram (the crosses signify the luminescence color coordinates) (b) of SA. Solvent effect on the light-emitting properties (c) and CIE 1931 chromaticity diagram (the crosses signify the luminescence color coordinates) (d) of SB. DMSO (dimethyl sulfoxide); MeOH (methanol); CH₃CN (acetonitrile); CP (acetone); and DCM (dichloromethane) ($\lambda_{\text{ex}} = 365 \text{ nm}$, $c = 10.0 \mu\text{M}$).

groups (Figure 3a). Compared with these ligands, their assemblies exhibited a significantly enhanced molar absorption coefficient because of the inclusion of multiple ligands in one metallo-cage structure (Figure 3b).⁴³ The absorption spectra of metallo-cage SA in DMSO exhibited the main absorption peaks at 290 and 360 nm originating mainly from pyridine and TPA. Similarly, metallo-cage SB in DMSO exhibited the main absorption peaks at 305 and 365 nm. As shown in Figure 3b, fluorescence emission spectra of metallo-cages SA and SB both exhibited an emission band at 455 nm, corresponding to the TPA group, which was bathochromically shifted by $\sim 25 \text{ nm}$ compared with ligand L owing to the metal-to-ligand charge transfer (MLCT) after coordination. It is worth mentioning

that the metallo-cage SB overcame the heavy atom effect of Pt(II) atoms by exhibiting an absolute quantum efficiency ($\Phi_{\text{F}} = 53.89\%$) higher than that of the ligand L ($\Phi_{\text{F}} = 40.68\%$) and also higher than that of metallo-cage SA ($\Phi_{\text{F}} = 31.99\%$). This is mainly attributed to the formation of metal–ligand bonds that hindered the intramolecular rotations of the triphenylamine core. Second, metallo-cage SB showed higher fluorescence emission because of its electron-absorbing end group (carbonyl), which formed a donor–acceptor interaction with triphenylamine.

We further probed their photophysical properties in a wide range of solvents, as shown in Figure 4a,c. Both metallo-cages SA and SB showed irregular color changes in solvents of

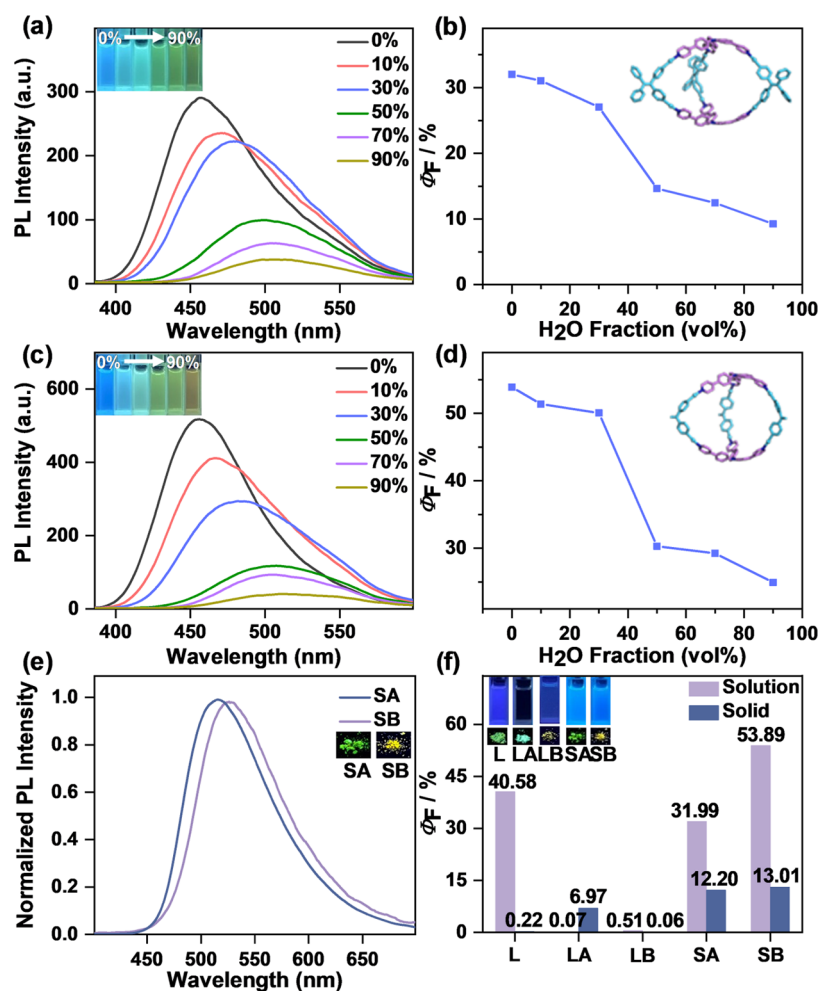


Figure 5. PL spectra of SA (a) and SB (c) recorded in DMSO/H₂O mixtures containing different H₂O fractions, respective inserted pictures show SA and SB dissolved in DMSO/H₂O from 0 to 90% H₂O. PLQYs of SA (b) and SB (d) versus increasing H₂O fractions in DMSO/H₂O mixtures. Fluorescence emission spectra of SA and SB in the solid state (e). PLQYs of ligands and metallo-cages in DMSO and in the solid state (f) (λ_{ex} = 365 nm, c = 10.0 μM).

different polarities, attributed to the effect of solvent polarity on intramolecular charge transfer. In polar aprotic solvents such as DMSO and acetonitrile, metallo-cages SA and SB show strong emission at around 455 nm due to the emission of the TPA groups. In less polar aprotic solvents of acetone, metallo-cages SA and SB show the main emission at around 470 nm. Also, in dichloromethane, metallo-cages SA and SB show two broad emission peaks at 430 and 500 nm, respectively. In contrast, in polar protic solvents (e.g., methanol), these metallo-cages show only a single emission peak at 505 nm. Furthermore, as can be seen from the CIE (Figure 4b,d), the change in the SA and SB fluorescence was also consistent with their fluorescence photographs. These data suggest that their luminous colors can be finely tuned by the variation of solvents.

To obtain further optical properties of these metallo-cages, the emission spectra in DMSO and DMSO/H₂O mixture solutions were recorded (Figure 5a–d). The emissive intensity and photoluminescent quantum yields (PLQYs) of SA and SB gradually decrease with increasing water content. Surprisingly, the PLQY decreases very slowly when the water content is below 30%. When the water content reached 90%, the PLQY of SA was still 9.26% and that of SB was 24.94%. In contrast, triphenylamine ligand L barely fluoresces when the water

content exceeds 70% (Φ_F = 1.1%) (Figure S25). This implied that the formation of metallo-cages can effectively hinder the π – π stacking between triphenylamine in the aggregated state. This finding prompted us to further explore the solid-state luminescence of metallo-cages SA and SB. As shown in Figure 5e, the maximum emission peak of the metallo-cage SB was red-shifted by 10 nm compared to that of SA. Under 365 nm UV light, SA and SB exhibited green and yellow fluorescence, respectively. The PLQY of fluorescence in dilute solution and solid state for all ligands and complexes is shown in Figure 5f. Ligand L showed extremely low luminescence efficiency (Φ_F = 0.22%) in the solid state due to molecular aggregation. However, the PLQYs of metallo-cages SA and SB exhibited a nearly 60 times improvement (12.20% for SA and 13.01% for SB), indicating that the curved conformation can suppress intermolecular stacking. It is worth noting that the absolute quantum efficiency of SA containing the TPE-based di-Pt(II) acceptor was only 2.6 times lower in the solid state compared to that in the dilute solution, while SB was 4.1 times lower. This is because the TPE moieties in SA can effectively compensate for part of the fluorescence quench. This also demonstrated that the metallo-cages combined the advantages of AIE and ACQ to achieve high-efficiency luminescence in both dilute solution and solid state.

CONCLUSIONS

In conclusion, we successfully constructed two metallo-cages **SA** and **SB** by using the TPA-based ligand **L** and 120° di-Pt(II) acceptors. The presence of metal coordination bonds caused a curved conformation of metallo-cages, which effectively restricted the free rotation of the benzene ring of the triphenylamine core. These metallo-cages with precise conformational control not only resulted in high emission in dilute solution but also enabled it to maintain strong fluorescence in the solid state due to the conformation of the metallo-cages reducing the π - π stacking of triphenylamine. This study will present a novel strategy for designing luminescent materials and also expand the range of luminescent MOCs.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.inorgchem.3c02054>.

Synthetic and characterization data obtained from NMR, ESI-MS, and photophysical studies (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Yang, S. K.; Shi, X.; Park, S.; Ha, T.; Zimmerman, S. C. A dendritic single-molecule fluorescent probe that is monovalent, photostable and minimally blinking. *Nat. Chem.* **2013**, *5* (8), 692–697.
- (2) Li, C.; Tang, Q.; Wei, H.; Liu, J.; Wang, Q.; Wang, Y.; Du, Z.; Wang, J.; Xu, R.; Bi, Y.; Snow, C. D.; Belfiore, L. A.; Tang, J. Smart Wearable Fluorescence Sensing of Bacterial Pathogens and Toxic Contaminants by Eu³⁺-Induced Sodium Alginate/Ag Nanoparticle Aggregates. *ACS Appl. Nano Mater.* **2022**, *5* (6), 8393–8403.
- (3) Zhang, Z.; Guo, K.; Li, Y.; Li, X.; Guan, G.; Li, H.; Luo, Y.; Zhao, F.; Zhang, Q.; Wei, B.; Pei, Q.; Peng, H. A colour-tunable, weavable fibre-shaped polymer light-emitting electrochemical cell. *Nat. Photonics* **2015**, *9* (4), 233–238.
- (4) Purba, P. C.; Venkateswaralu, M.; Bhattacharyya, S.; Mukherjee, P. S. Silver(I)–Carbene Bond-Directed Rigidification-Induced Emissive Metallacage for Picric Acid Detection. *Inorg. Chem.* **2022**, *61* (1), 713–722.
- (5) Tian, W.; Zhang, J.; Yu, J.; Wu, J.; Nawaz, H.; Zhang, J.; He, J.; Wang, F. Cellulose-Based Solid Fluorescent Materials. *Adv. Opt. Mater.* **2016**, *4* (12), 2044–2050.
- (6) Tagare, J.; Vaidyanathan, S. Recent development of phenanthroimidazole-based fluorophores for blue organic light-emitting diodes (OLEDs): an overview. *J. Mater. Chem. C* **2018**, *6* (38), 10138–10173.
- (7) Leung, C. W. T.; Hong, Y.; Chen, S.; Zhao, E.; Lam, J. W. Y.; Tang, B. Z. A Photostable AIE Luminogen for Specific Mitochondrial Imaging and Tracking. *J. Am. Chem. Soc.* **2013**, *135* (1), 62–65.
- (8) Zhang, W.; Kwok, R. T. K.; Chen, Y.; Chen, S.; Zhao, E.; Yu, C. Y. Y.; Lam, J. W. Y.; Zheng, Q.; Tang, B. Z. Real-time monitoring of the mitophagy process by a photostable fluorescent mitochondrion-specific bioprobe with AIE characteristics. *Chem. Commun.* **2015**, *51* (43), 9022–9025.
- (9) Hoebe, F. J. M.; Jonkheijm, P.; Meijer, E. W.; Schenning, A. P. H. J. About Supramolecular Assemblies of π -Conjugated Systems. *Chem. Rev.* **2005**, *105* (4), 1491–1546.
- (10) Tang, C. W.; VanSlyke, S. A. Organic electroluminescent diodes. *Appl. Phys. Lett.* **1987**, *51* (12), 913–915.
- (11) Liu, Y.; Chen, S.; Lam, J. W. Y.; Lu, P.; Kwok, R. T. K.; Mahtab, F.; Kwok, H. S.; Tang, B. Z. Tuning the Electronic Nature of Aggregation-Induced Emission Luminogens with Enhanced Hole-Transporting Property. *Chem. Mater.* **2011**, *23* (10), 2536–2544.
- (12) Naito, H.; Morisaki, Y.; Chujo, Y. o-Carborane-Based Anthracene: A Variety of Emission Behaviors. *Angew. Chem., Int. Ed.* **2015**, *54* (17), 5084–5087.
- (13) Wu, X.; Guo, J.; Quan, Y.; Jia, W.; Jia, D.; Chen, Y.; Xie, Z. Cage carbon-substitute does matter for aggregation-induced emission features of o-carborane-functionalized anthracene triads. *J. Mater. Chem. C* **2018**, *6* (15), 4140–4149.
- (14) Kumar, A.; Seok Chae, P. Pyreneamide-based dipodal probes for ultra-sensitive and selective detection of 3,5-dinitrosalicylic acid in an aqueous solution. *Dyes Pigm.* **2017**, *147*, 400–412.
- (15) Islam, M. M.; Hu, Z.; Wang, Q.; Redshaw, C.; Feng, X. Pyrene-based aggregation-induced emission luminogens and their applications. *Mater. Chem. Front.* **2019**, *3* (5), 762–781.
- (16) Zhang, B.; Lyskov, I.; Wilson, L. J.; Sabatini, R. P.; Manian, A.; Soleimaninejad, H.; White, J. M.; Smith, T. A.; Lakhwani, G.; Jones, D. J.; Ghiggino, K. P.; Russo, S. P.; Wong, W. W. H. FRET-enhanced photoluminescence of perylene diimides by combining molecular aggregation and insulation. *J. Mater. Chem. C* **2020**, *8* (26), 8953–8961.

- (17) Luo, J.; Xie, Z.; Lam, J. W. Y.; Cheng, L.; Chen, H.; Qiu, C.; Kwok, H. S.; Zhan, X.; Liu, Y.; Zhu, D.; Tang, B. Z. Aggregation-induced emission of 1-methyl-1,2,3,4,5-pentaphenylsilole. *Chem. Commun.* **2001**, No. 18, 1740–1741.
- (18) Li, M.; Jiang, S.; Zhang, Z.; Hao, X.-Q.; Jiang, X.; Yu, H.; Wang, P.; Xu, B.; Wang, M.; Tian, W. Tetraphenylethylene-Based Emissive Supramolecular Metallacages Assembled by Terpyridine Ligands. *CCS Chem.* **2020**, 2 (4), 337–348.
- (19) Shi, J.; Li, M.; Su, H.; Bai, Q.; Han, N.; Hao, X.-Q.; Fang, F.; Zhang, Z.; Wang, P.; Ma, B.; Wang, M. Construction of metallo-triangles with cis-TPE motifs and fluorescence properties. *Chem. Commun.* **2022**, 58 (99), 13767–13770.
- (20) Nishino, K.; Yamamoto, H.; Tanaka, K.; Chujo, Y. Solid-State Thermochromic Luminescence through Twisted Intramolecular Charge Transfer and Excimer Formation of a Carborane–Pyrene Dyad with an Ethynyl Spacer. *Asian. J. Org. Chem.* **2017**, 6 (12), 1818–1822.
- (21) Yang, G.; Tang, Y.; Li, X.; Ågren, H.; Xie, Y. Efficient Solar Cells Based on Porphyrin Dyes with Flexible Chains Attached to the Auxiliary Benzothiadiazole Acceptor: Suppression of Dye Aggregation and the Effect of Distortion. *ACS Appl. Mater. Interfaces* **2017**, 9 (42), 36875–36885.
- (22) Hamzehpoor, E.; Ruchlin, C.; Tao, Y.; Liu, C.-H.; Titi, H. M.; Perepichka, D. F. Efficient room-temperature phosphorescence of covalent organic frameworks through covalent halogen doping. *Nat. Chem.* **2023**, 15 (1), 83–90.
- (23) Sáringer, S.; Akula, R. A.; Szerlauth, A.; Szilagyi, I. Papain Adsorption on Latex Particles: Charging, Aggregation, and Enzymatic Activity. *J. Phys. Chem. B* **2019**, 123 (46), 9984–9991.
- (24) Leininger, S.; Olenyuk, B.; Stang, P. J. Self-Assembly of Discrete Cyclic Nanostructures Mediated by Transition Metals. *Chem. Rev.* **2000**, 100 (3), 853–908.
- (25) Chakraborty, S.; Newkome, G. R. Terpyridine-based metallosupramolecular constructs: tailored monomers to precise 2D-motifs and 3D-metallacages. *Chem. Soc. Rev.* **2018**, 47 (11), 3991–4016.
- (26) Chan, Y.-T.; Li, X.; Soler, M.; Wang, J.-L.; Wesdemiotis, C.; Newkome, G. R. Self-Assembly and Traveling Wave Ion Mobility Mass Spectrometry Analysis of Hexacadmium Macrocycles. *J. Am. Chem. Soc.* **2009**, 131 (45), 16395–16397.
- (27) Xu, Y.; Yu, H.; Jiang, X.; Shi, J.; Li, B.; Li, L.; Wu, L.; Wang, M. Porous Assembly of Metallo-Supramolecule and Polyoxometalate via Ionic Complexation with Vapor Sorption Properties. *Chin. J. Chem.* **2022**, 40 (7), 813–818.
- (28) Siddiqui, M. M.; Saha, R.; Mukherjee, P. S. Ruthenium(II) Metalla[2]catenanes and Macrocycles via Donor-Dependent Self-Assembly. *Inorg. Chem.* **2019**, 58 (7), 4491–4499.
- (29) Yu, H.; Shi, J.; Li, M.; Pan, G.; Tong, H.; Xu, B.; Wang, M.; Tian, W. Discrete Platinum(II) Metallacycles with Inner- and Outer-Modified 9,10-Distyrylanthracene: Design, Self-Assembly, and Luminescence Properties. *Inorg. Chem.* **2022**, 61 (19), 7231–7237.
- (30) Tang, J.-H.; Ni, R.; He, Y.-Q.; Vanderlinden, R. T.; Li, Y.; Shi, B.; Li, Z.-Y.; Wang, H.; Li, X.; Sun, Y.; Zhong, Y.-W.; Stang, P. J. Metal–Organic Pt(II) Hexagonal-Prism Macrocycles and Their Photophysical Properties. *Inorg. Chem.* **2019**, 58 (19), 13376–13381.
- (31) Jiang, X.; Yu, H.; Shi, J.; Bai, Q.; Xu, Y.; Zhang, Z.; Hao, X.-Q.; Li, B.; Wang, P.; Wu, L.; Wang, M. From Mechanically Interlocked Structures to Host–Guest Chemistry Based on Twisted Dimeric Architectures by Adjusting Space Constraints. *CCS Chem.* **2021**, 2042–2054.
- (32) Zeng, Y.; Jiang, X.; Yu, H.; Cao, Y.; Lai, Z.; Shi, J.; Wang, M. Coordination-Driven Terpyridine-Based Twisted Prisms with Tunable Emissions and Hierarchical Self-Assembly Properties. *Adv. Opt. Mater.* **2022**, 10 (10), No. 2102613.
- (33) Howlader, P.; Ahmed, S.; Mondal, S.; Zangrando, E.; Mukherjee, P. S. Conformation-Selective Self-Assembly of Pd6 Trifacial Molecular Barrels Using a Tetrapyridyl Ligand. *Inorg. Chem.* **2022**, 61 (21), 8121–8125.
- (34) Mu, C.; Zhang, Z.; Hou, Y.; Liu, H.; Ma, L.; Li, X.; Ling, S.; He, G.; Zhang, M. Tetraphenylethylene-Based Multicomponent Emissive Metallacages as Solid-State Fluorescent Materials. *Angew. Chem., Int. Ed.* **2021**, 60 (22), 12293–12297.
- (35) Zhao, Z.; Liu, J.; Yip Lam, J. W.; Chan, C. Y. K.; Qiu, H.; Tang, B. Z. Luminescent aggregates of a starburst silole-triphenylamine adduct for sensitive explosive detection. *Dyes Pigm.* **2011**, 91 (2), 258–263.
- (36) Wu, B.; Guo, Z.; Li, G.; Zhao, J.; Liu, Y.; Wang, J.; Wang, H.; Yan, X. Synergistic combination of ACQ and AIE moieties to enhance the emission of hexagonal metallacycles. *Chem. Commun.* **2021**, 57 (84), 11056–11059.
- (37) Yan, X.; Wang, M.; Cook, T. R.; Zhang, M.; Saha, M. L.; Zhou, Z.; Li, X.; Huang, F.; Stang, P. J. Light-Emitting Superstructures with Anion Effect: Coordination-Driven Self-Assembly of Pure Tetraphenylethylene Metallacycles and Metallacages. *J. Am. Chem. Soc.* **2016**, 138 (13), 4580–4588.
- (38) Zhou, Z.; Yan, X.; Saha, M. L.; Zhang, M.; Wang, M.; Li, X.; Stang, P. J. Immobilizing Tetraphenylethylene into Fused Metallacycles: Shape Effects on Fluorescence Emission. *J. Am. Chem. Soc.* **2016**, 138 (40), 13131–13134.
- (39) Chan, Y.-T.; Li, X.; Yu, J.; Carri, G. A.; Moorefield, C. N.; Newkome, G. R.; Wesdemiotis, C. Design, Synthesis, and Traveling Wave Ion Mobility Mass Spectrometry Characterization of Iron(II)– and Ruthenium(II)–Terpyridine Metallomacrocycles. *J. Am. Chem. Soc.* **2011**, 133 (31), 11967–11976.
- (40) Wang, M.; Wang, C.; Hao, X.-Q.; Li, X.; Vaughn, T. J.; Zhang, Y.-Y.; Yu, Y.; Li, Z.-Y.; Song, M.-P.; Yang, H.-B.; Li, X. From Trigonal Bipyramidal to Platonic Solids: Self-Assembly and Self-Sorting Study of Terpyridine-Based 3D Architectures. *J. Am. Chem. Soc.* **2014**, 136 (29), 10499–10507.
- (41) Frisch, M. J.; Trucks, G.; Schlegel, H.; Scuseria, G.; Robb, M.; Cheeseman, J.; Scalmani, G.; Barone, V.; Mennucci, B.; Petersson, G.; Nakatsuji, H.; Caricato, M.; Li, X.; Hratchian, H.; Izmaylov, A.; Bloino, J.; Zheng, G.; Sonnenberg, J.; Hada, M.; Fox, D. *Gaussian 16*; Gaussian, Inc.: Wallingford CT, 2004.
- (42) Grimme, S.; Antony, J.; Ehrlich, S.; Krieg, H. A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H–Pu. *J. Chem. Phys.* **2010**, 132 (15), No. 154104.
- (43) Tang, J.-H.; Sun, Y.; Gong, Z.-L.; Li, Z.-Y.; Zhou, Z.; Wang, H.; Li, X.; Saha, M. L.; Zhong, Y.-W.; Stang, P. J. Temperature-Responsive Fluorescent Organoplatinum(II) Metallacycles. *J. Am. Chem. Soc.* **2018**, 140 (24), 7723–7729.